PRODUCT EXPERIMENT / DECISION MEMO

Should we launch the new checkout?

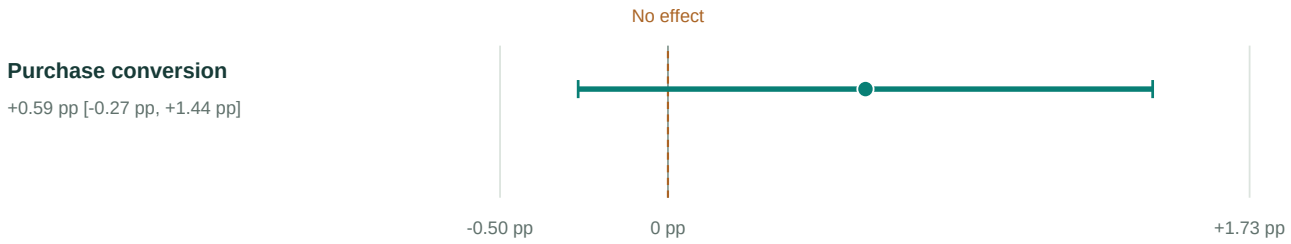
24,000 fictional users. One accountable launch decision.

CONTINUE TESTING - CONDITIONAL ON FEASIBILITY

Conversion is promising; revenue and refund safety remain unresolved.

Decision: Retain the existing checkout. Treat this as an inconclusive pilot and assess whether a fresh, independently planned test is economically feasible.

01 / Does the new checkout improve conversion?



Control: 12.86%. New: 13.45%. Relative lift: +4.57% (95% CI -2.01% to +11.60%). The absolute interval includes no improvement.

02 / Do revenue and refunds clear the safety criteria?

Metric	Difference [95% CI]	Safety
Gross revenue / user	-$0.06 [-$1.45 to $1.27]	Secondary
After-refund revenue / user	-$0.28 [-$1.59 to $0.98]	Unresolved
Refunded-user rate	+0.21 pp [-0.08 pp to +0.51 pp]	Unresolved

Illustrative launch rule: conversion lower 95% bound > 0; after-refund revenue/user lower bound > -$0.25; refund-rate upper bound < +0.30 pp. All must pass. Current outcome criteria cleared: 0 of 3. Margins require stakeholder agreement before confirmation.

Assignment checks: 11,911 control / 12,089 new; SRM p=0.251. 24 exact ingestion retries removed, no assignment conflicts, and all 20,842 nonbuyers retained.

More buyers, nearly flat revenue estimate

Gross revenue per assigned user is -$0.06 at the point estimate, with a 95% interval from -$1.45 to $1.27. The bridge is descriptive and does not establish separate causal mechanisms.

More buyers	**+$0.66**
Orders per buyer	**-$0.19**
Average order value	**-$0.53**
Additional refunds	**-$0.22**
Total net difference	**-$0.28**

USD per assigned user | exact descriptive decomposition

03 / The next test is constrained by revenue precision

Conversion alone needs approximately 49,838 users per arm to detect +0.60 percentage points with 80% power. The $0.25 revenue-loss safety margin needs about 644,548 per arm even assuming zero true loss. At 50,000 per arm, that safety check has only 12% approximate power.

Agree on acceptable economic loss, available traffic, test cost and sample cap before starting. If the required precision is unaffordable, retain the current checkout and redesign the proposal. Do not relax a safety margin merely to make the test pass.

Measurement and interpretation

1. Intention-to-treat: all assigned users. Paid orders in [assignment, +7 days); refunds on eligible orders in [order, +30 days). All follow-up is mature.

2. 95% proportion intervals: Newcombe/Wilson. Revenue: 10,000 within-arm user bootstrap resamples, including nonbuyers and repeat orders.

3. Planning figures are approximate marginal powers, not a joint launch guarantee. Further testing uses a fresh fixed-horizon run; avoid optional stopping.

4. No segment wins are claimed. Multiple subgroup searches can create false discoveries; use planned interactions and multiplicity control.

5. Synthetic data do not validate a real retailer. After-refund revenue excludes costs, fees and long-term effects; it is not profit.